

BAHLA, Oman

WMO: 412630

Lat: 22.9122N

Lon: 57.2572E

Elev: 1683

StdP: 13.82

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.5	55.2	15.1	12.6	84.2	19.3	15.4	82.1	22.9	72.7	19.6	72.9	1.9	0	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.8	111.5	67.8	109.6	68.1	107.8	68.1	77.3	91.4	76.2	91.5	75.3	91.7	10.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	133.9	84.6	72.1	126.6	84.5	70.7	120.4	84.7	42.3	90.8	41.1	91.3	40.1	91.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.1	14.0	DB	48.2	115.0	2.4	1.3	46.5	116.0	45.1	116.8	43.7	117.5	41.9	118.5	
			WB	38.5	78.1	3.3	1.6	36.1	79.2	34.2	80.2	32.3	81.1	29.9	82.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	66.6	70.7	77.5	85.7	93.1	96.4	95.8	93.8	90.4	84.7	75.7	69.1
	DBStd	11.18	3.55	4.36	4.52	4.28	3.48	3.81	4.13	4.09	3.13	3.24	3.33	3.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	41	26	5	0	0	0	0	0	0	0	0	0	10
	CDD50	12172	514	579	852	1071	1335	1392	1421	1358	1211	1076	770	593
	CDD65	6738	75	164	387	621	870	942	956	893	761	611	320	138
	CDH74	100937	494	1474	4431	8868	14438	16152	16151	14352	11886	8530	3156	1005
CDH80	65194	53	394	2021	5232	10021	11846	11730	9991	7749	4892	1096	169	
Wind	WSAvg	5.5	4.5	5.3	5.8	6.1	6.8	7.1	6.3	5.7	5.6	5.1	4.2	4.0
Precipitation	PrecAvg	4.6	0.3	0.7	0.6	0.7	0.3	0.3	0.6	0.7	0.3	0.3	0.2	0.1
	PrecMax	8.1	2.5	4.1	3.4	3.4	1.3	1.3	1.6	1.9	0.8	1.9	1.6	0.8
	PrecMin	1.8	0.0	0.2	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.6	0.9	0.9	1.0	0.4	0.4	0.4	0.5	0.2	0.5	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.8	90.5	97.1	103.4	112.3	114.8	113.3	111.4	106.6	102.2	92.6	86.6
		MCWB	57.7	61.0	61.2	64.3	65.9	67.4	69.1	69.4	67.5	64.7	63.3	59.9
	2%	DB	81.4	87.3	94.2	100.6	108.1	112.2	110.9	109.0	104.5	99.7	89.9	83.9
		MCWB	58.1	60.0	60.8	63.7	66.3	66.9	68.7	69.3	67.2	64.0	62.0	59.2
	5%	DB	79.0	84.6	91.8	98.7	105.7	109.8	109.0	107.0	102.8	97.8	88.1	81.8
		MCWB	57.9	59.4	61.1	63.3	65.9	67.0	69.0	69.3	67.0	63.7	62.0	59.5
	10%	DB	76.7	82.0	89.5	96.7	103.6	107.7	107.1	104.9	101.0	95.7	86.0	79.5
		MCWB	57.6	58.9	60.6	62.4	65.5	67.2	69.2	69.9	67.2	63.2	61.4	59.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.8	67.5	70.1	72.0	76.7	79.0	78.3	77.9	75.6	72.9	71.8	66.1
		MCDB	73.7	77.6	79.9	82.5	85.8	89.8	94.6	92.1	91.5	85.1	81.0	72.3
	2%	WB	62.7	65.4	68.1	69.5	74.7	77.5	77.0	76.1	74.5	71.0	67.6	64.3
		MCDB	71.9	77.1	79.2	83.0	87.3	90.2	93.1	92.0	90.2	84.0	79.0	74.4
	5%	WB	61.6	63.8	66.5	67.9	73.3	76.1	76.1	75.1	73.5	69.5	65.8	63.3
		MCDB	71.8	76.6	80.2	85.1	88.7	90.7	92.6	93.2	88.8	84.3	78.9	74.6
	10%	WB	60.3	62.0	64.9	66.4	71.6	74.7	75.2	74.3	72.6	68.1	64.6	62.3
		MCDB	71.7	75.8	79.8	88.5	90.4	91.4	92.2	92.6	88.4	85.0	78.9	74.5
Mean Daily Temperature Range	5% DB	MDBR	21.1	21.9	21.9	20.5	20.8	21.8	21.3	21.6	21.6	21.5	20.3	20.6
		MCDBR	23.3	24.5	24.1	21.7	22.6	23.7	22.4	22.9	22.5	23.0	21.6	23.2
		MCWBR	8.9	9.1	8.4	6.7	7.3	7.5	6.1	5.5	6.2	7.1	8.0	8.8
	5% WB	MCDBR	18.2	18.8	17.9	19.0	20.1	22.4	21.3	22.0	21.5	19.8	18.3	18.4
		MCWBR	7.6	7.9	7.5	7.1	8.6	8.0	5.2	4.8	5.8	6.7	7.0	7.7
Clear-Sky Solar Irradiance	taub		0.424	0.480	0.556	0.626	0.662	0.741	0.816	0.777	0.658	0.516	0.453	0.421
	taud		2.087	1.918	1.733	1.627	1.568	1.467	1.396	1.443	1.601	1.854	2.016	2.088
	Ebn at Noon		262	254	241	227	218	200	186	193	213	240	250	257
	Edh at Noon		47	59	74	83	88	97	104	99	83	62	49	45
All-Sky Solar Radiation	RadAvg		1503	1753	2023	2228	2411	2358	2175	2154	2050	1873	1570	1424
	RadStd		62	73	105	116	50	84	87	52	38	50	37	58

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	+0.76	+1.22	N/A	N/A	+0.50	+0.54	N/A	N/A	+173	+176	

Nomenclature: See separate page